Software Dependencies and Requirements

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Pipeline Component Dependencies

Installation Order

1. Install NVIDIA Drivers
   https://www.nvidia.com/download/index.aspx

2. Install CUDA Toolkit
   https://developer.nvidia.com/cuda-downloads

3. Install Microsoft Visual C++ Redistributable
   https://learn.microsoft.com/en-us/cpp/windows/latest-supported-vc-redist

4. Install Python 3.11
   https://www.python.org/downloads/

5. Install COLMAP
   https://colmap.github.io/install.html

6. Install Brush
   https://github.com/ArthurBrussee/brush

7. Create a Python virtual environment

8. Install PyTorch and Torchvision
   https://pytorch.org/get-started/locally/

9. Install NumPy, Pillow, OpenCV and Transformers
   NumPy: https://numpy.org/install/
   Pillow: https://pillow.readthedocs.io/en/stable/installation.html

OpenCV: https://opencv.org/
Transformers: https://huggingface.co/docs/transformers

10. Run the pipeline

Stage 1 – COLMAP Reconstruction

Dependencies

- Python 3.11
- Pillow
- COLMAP
- CUDA-enabled COLMAP (optional)
- NVIDIA GPU (optional)

Stage 2 – Brush Gaussian Splatting

Dependencies

- Brush
- CUDA Toolkit
- NVIDIA GPU
- Microsoft Visual C++ Redistributable

Stage 3 – Semantic Segmentation, Object Detection and 3D Semantic Lifting

Dependencies

- NumPy
- Pillow
- OpenCV
- PyTorch
- Torchvision
- Transformers
- CUDA-enabled PyTorch (recommended)

- SegFormer
- Grounding DINO
- Segment Anything Model (SAM)

Stage 3 Components

Semantic Segmentation

- SegFormer semantic segmentation model

Object Detection

- Grounding DINO

Mask Generation

- Segment Anything Model (SAM)

2D to 3D Semantic Lifting

- Vote-based semantic transfer from 2D image labels to 3D Gaussian splats

Operating System

- Windows 10 (64-bit)
- Windows 11 (64-bit)

Hardware

Recommended Specifications

- NVIDIA RTX-series GPU
- Minimum 8 GB VRAM
- Preferred 12 GB+ VRAM

NVIDIA Drivers

Install the latest NVIDIA graphics drivers.

Verify Installation

nvidia-smi

CUDA Toolkit

Required when using GPU acceleration.

Recommended Version

- CUDA Toolkit 12.4

Verify Installation

nvcc --version

or

nvidia-smi

Microsoft Visual C++ Redistributable

Install

- Microsoft Visual C++ Redistributable 2015–2022 (x64)

Required By

- COLMAP
- PyTorch
- Brush

External Software

Python

Recommended

- Python 3.11

Avoid

- Python 3.13 (limited support for some packages)

COLMAP

Required for Structure-from-Motion reconstruction.

Recommended Version

- COLMAP 3.9 or newer

Two options are available:

1. **COLMAP CPU Build**
2. **COLMAP CUDA Build (recommended)**

Uses the GPU for:

- Feature extraction
- Feature matching

The GPU wrapper script (Script_S1_Colmap_GPU.py) requires a CUDA-enabled COLMAP build.

<u>Brush</u>

Required for training the Gaussian Splat model.

The Brush executable must be:

- Added to the system PATH, or
- Specified using:

--brush_path <path_to_brush.exe>

<u>Python Dependencies</u>

Install PyTorch first.

CUDA-enabled PyTorch

For CUDA 12.4:

pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu124

Additional Packages

pip install numpy pillow opencv-python transformers

<u>Python Package Versions</u>

Recommended versions:

numpy>=1.26

pillow>=10.0

opencv-python>=4.9

transformers>=4.40

torch>=2.4

torchvision>=0.19

torchaudio>=2.4

AI Model Dependencies

The semantic processing stage automatically downloads the required pretrained models during first execution.

SegFormer

Used for semantic segmentation of structural indoor elements.

Model:

nvidia/segformer-b0-finetuned-ade-512-512

Grounding DINO

Used for text-prompted object detection of wayfinding-related objects such as signs.

Model:

IDEA-Research/grounding-dino-base

Segment Anything Model (SAM)

Used to generate pixel-level masks from Grounding DINO detections.

Model:

facebook/sam-vit-base

Internet Requirement

An internet connection is required during the first execution to download the pretrained models. After downloading, the models are stored locally and can be reused offline.